EVALUATION OF WATER QUALITY INDEX FOR RIVER MOGRAL

A PROJECT REPORT

submitted by

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

In

Civil Engineering

DEPARTMENT OF CIVIL ENGINEERING

LBS COLLEGE OF ENGINEERING KASARAGOD

KASARAGOD-671542, KERALA

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DECLARATION

We under signed hereby declare that the project report–Evaluation of Water Quality Index for River Mogral", submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by we under supervision of Mrs. Anjali MS. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission.We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained.This report has not been previously formed the basis for the award of any degree, diplomaor similar title of any other University.

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CERTIFICATE

This is to certify that the project report entitled **"EVALUATION OF WATER QUALITY INDEX FOR RIVER MOGRAL"** submitted by **Abdul Faeem, Ahammed Misab CV, Ayishath Rusaina, Fathima AB'** to the APJ Abdul Kalam KERALA Technological University in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in Civil Engineering is a bonafide record of work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

The present study was carried out to evaluate the physico-chemical parameter of water quality and to compute water quality index of Mogral river. The Mogral Riveris a west-flowing river in the Kasaragod district of Kerala having a length of 34 kilometres and a drainage area of 132 (119.997) square kilometres.

Water samples were collected from 24 different stations along the Mogral basin from Jan 2021 to Feb 2021.These samples were analyzed for six water parameters pH, chloride, alkalinity, Biological Oxygen Demand (BOD), Dissolved Oxygen (DO) and Temperature. The results of the analysis were compared with the permissible limits prescribed by Bureau of Indian Standard (BIS). Water Quality Index has been determined by using Weighted Arithmetic Water Quality Index method proposed by Horton in 1965 and the water quality status of each sample were found out. The analysis showed that the chloride concentration was higher than the limits prescribed by BIS standards. This indicated saline water intrusion into the river. The majority of the sites fall under good category and some sites were not found to be in the desirable quality. The lower quality at the sites revealed local pollution due to anthropogenic activities and entry of sewage into the river. Hence appropriate measures were suggested to improve the quality of water.

KEYWORDS: Physico-Chemical parameter, Water Quality Index (WQI), Bureau of Indian Standard (BIS), Dissolved Oxygen (DO), Biological Oxygen Demand (BOD)

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LIST OF ABBREVIATIONS

BCWQI : BRITISH COLUMBIA WATER QUALITY INDEX

BIS : BUREAU OF INDIAN STANDARD

BOD : BIOLOGICAL OXYGEN DEMAND

CCMEWQI: CANADIAN COUNCIL OF MINISTERS OF THE ENVIRONMENT WATER QUALITY INDEX

CPCB : CENTRAL POLLUTION CONTROL BOARD

DO : DISSOLVED OXYGEN

ICMR : INDIAN COUNCIL OF MEDICAL RESEARCH

NSFWQI : NATIONAL SANITATION FOUNDATION WATER QUALITY INDEX

WAWQI : WEIGHTED ARITHMATIC WATER QUALITY INDEX

WHO : WORLD HEALTH ORGANIZATION

WQI : WATER QUALITY INDEX

CHAPTER 1

INTRODUCTION

For the human and industrial growth, water is considered to be the main requirement. Increase in population and industrialization, the demand of the freshwater increases in the last decades. This demand fulfilled by the rivers which provide the water for human life and agriculture purposes. Due to the waste discharged from the human and industrial activities, the quality of river water has deteriorated which affects human as well as aquatic life. According to WHO, CPCB, BIS, ICMR, the water quality of about 70% river water was contaminated due to pollutants in India and some of the river water was too poor for human.

Assessment of quality of river water using various parameters (physico-chemical and biological) and the different ways and techniques to protect the river water have been reported in the literature (Santosh et al., 2008; Yisa and Jimoh, 2010; Shah et al., 2015). One approach for determination of quality of river water is water quality index, found to be an efficient and useful method for assessing the water quality. This method gives an idea about the overall quality of water to the concern policy makers. Water quality index incorporate the different physical, chemical and biological parameters for the determination of water quality indices using the various mathematical equations.

The use of a WQI was initially proposed by Horton (1965) and Brown et al(1970). Since then, many different methods for the calculation of WQI's have been developed. The different water quality index used worldwide are US National Sanitation Foundation Water Quality Index (NSFWQI), Canadian Council of Ministers of the Environment Water Quality Index (CCMEWQI), British Columbia Water Quality Index (BCWQI), Oregon Water Quality Index (OWQI), Weighted Arithmetic Water Quality Index (WAWQI). Generally two steps are needed for the calculation of water quality indices. First one is the transformation of the selected water quality characteristics into sub index values. Second is the aggregation of these values for the water quality index value.

Studies on the effect of the various parameters on the quality of water of the Mogral River has been performed and discuss the suitability of the water for human consumption based on values of water quality index. The method used here for the calculation of WQI is Weighted Arithmetic Water Quality Index (WAWQI).

1.1 STUDY AREA

Mogral Puthur is a gram panchayat in the Kasaragod District, state of Kerala, India. It is to north of the Kasaragod, and it consists of two villages; Puthur and Kudlu. Mogral Puthur has on its west border the Arabian Sea; on the east is the Madhur panchayat; and on the north is the Mogral River.

The Mogral Puthur topography ranges from small hills and slopes to flat plains. The Mogral River, which forms its northern and a large part of the eastern border, supports much of its agriculture. There are several small monsoon-fed rivulets throughout the panchayat. Low-lying areas close to the U-shaped river and its tributaries are very fertile. The land gains elevation in the central part of the panchayat. In the central and eastern parts, there are rich laterite deposits.

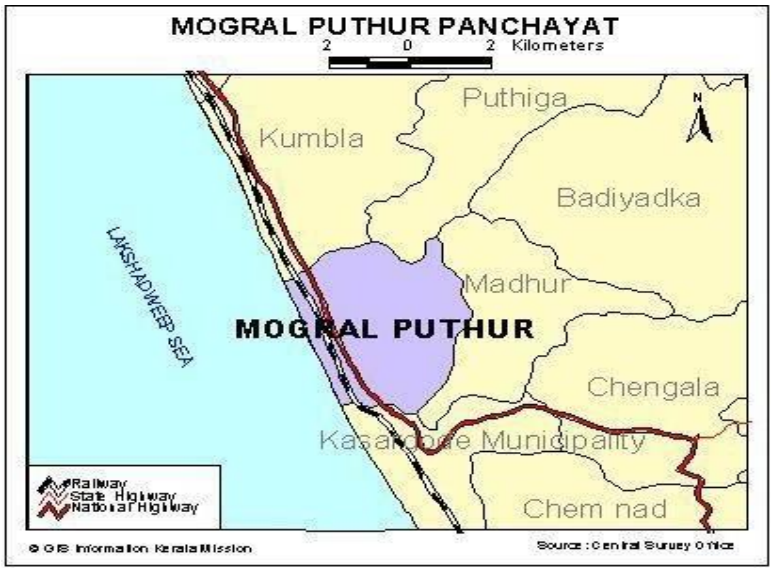

Fig no 1.1 MogralPuthur

1.1.1 Sources of pollution

- Sand Mining Activities

River sand is preferred for construction because it requires less processing and has better quality than other sources. But it comes at a huge cost to the river and those living around it. Excessive sand mining can alter the river bed, force the river to change course, erode banks and lead to flooding. It also destroys the habitat of aquatic animals and micro-organisms besides affecting groundwater recharge

Fig no 1.2 Sand mining activities

- Agricultural Runoff

Agriculture, which accounts for 70 percent of water withdrawals worldwide, plays a major role in water pollution. Farms discharge large quantities of agrochemicals, organic matter, drug residues, sediments and saline drainage into water bodies. Runoff from agricultural land can carry excess nutrients, such as nitrogen and phosphorus into streams, lakes, and groundwater supplies. These excess nutrients have the potential to degrade water quality.

Fig no 1.3 Agricultural runoff

- Domestic wastes

Domestic wastes like paper , cardboard , wood , fruits and vegetables peels and sand etc can have chemical and biological impacts on rivers and streams. Among the many impacts are

interfering with the establishment of aquatic plants,affecting the reproductive behavior of fish and other animals, and depleting the water of dissolved oxygen as the wastes decompose.

Fig no 1.4 Domestic wastes

- Bathing and Washing

The soap and detergent used in bathing or washing contain certain chemicals which can pollute the rivers .Therefore leading to water pollution.It also results in death of marine life.

Fig no 1.5 Bathing and washing

1.2 GEOMORPHOLOGY OF THE AREA

The district can be divided into three physiographical parts viz. the coastal plains, the midlands and the eastern highland regions. The coastal plains with an elevation of less than 10m occur as narrow belt of alluvial deposits parallel to the coast. To the east of coastal belt is the midland region with altitude ranging from 10 to 300m above mean sea level. The midland area is characterized by rugged topography formed by small hillocks separated by deep cut valleys. The midland regions show a general slope towards the western coast. To its east is the high land region. The midland and hill ranges of the district present a rugged and rolling topography with hills and valleys. Along the midlands the hills are mostly laterite and the valley are covered by valley fill deposits. The valley fill deposits are composed of colluvium and alluvium.

1.3 SOIL TYPES

There are four major soil types encountered in the district. They are Lateritic Soil, Brown Hydromorphic Soil, Alluvial Soil and Forest Loam. Lateritic soil is the most predominant soil type of the district and it occurs in the midland and hilly areas and it is derived from laterites. Brown hydromorphic soil is confined to the valleys between undulating topography in the midlands and in the low lying areas of the coastal strip. They have been formed as a result of transportation and sedimentation of materials from adjoining hill slopes. The alluvial soil is seen in the western coastal tract of the district. The coastal plain is characterized by secondary soils which are sandy and sterile with poor water holding capacity. The width of the zone increases towards the southern part of the district. Forest loamy soil is found in the eastern hilly areas of the district and are characterized by a surface layer rich in organic matter.

1.4 HYDROGEOLOGY

Four hydrogeological units encountered are alluvium, laterites, weathered crystalline, and fractured crystalline. Coastal Alluvium occurs as narrow strips parallel to the coast south of Kasaragod. The width of alluvium increases to the south and attains about 5 km around Trikaripur. North of Kasaragod (in Kasaragod and Manjeshwar blocks), the alluvium occurs as isolated patches close to the coast and have limited thickness. In the Kanhangad and Nileshwar blocks even though the width of alluvium is more, potential zones are seen in the top portion

only followed by tertiary sediments at deeper levels which does not contain potential granular zones. Valley fills occur in between laterite hills which are composed of colluvium and alluvium. The water level ranges from 2.93 m to 5.63 m below ground level in pre-monsoon period and 1.20 to 3.20 m below ground level in post monsoon period. The water level fluctuation is in the range of 0.98 to 2.68 m.

1.5 RAINFALL & CLIMATE

The district receives an average of about 3500 mm rainfall annually. The major source of rainfall is southwest monsoon from June to September which contributes nearly 85.3% of the total rainfall of the year. The northeast monsoon contributes nearly 8.9% and balance of 5.8% is received during the month of January to May as pre monsoon showers. Out of the 106 rainy days in a year, 87 rainy days occur during south west monsoon season.

1.6 METEOROLOGICAL PARAMETERS

• Temperature

The temperature is more during the months of March to May and is less during December and January. The average mean monthly maximum temperature ranges from 29.2 to 33.4∘C and minimum temperature ranges from 19.7 to 25∘C.

• Relative Humidity

Relative humidity is more during morning hours and is less during evening hours. During the morning hours it ranges from 87.1 to 98.7% and during evening hours it ranges from 54.4 to 86.5%.

• Evaporation

Evaporation is more during summer months of March to May. In general it ranges from 2.2 to 6.3 mm/day. During south west monsoon season it ranges from 2.2 to 3.7 mm/day.

• Sunshine Hours

Sunshine ranges from 3.2 to 10.2 hours/day. Maximum sunshine is during the month of February. The months of June to August record the minimum sunshine due to the cloudy sky. Generally good sunshine hours are recorded in the months of November to May.

• Wind

The wind speed ranges from 2.1 to 3.3 km/hour. The wind speed is high during the months of March to June and less during the months of September to December.

• Potential Evapotranspiration (PET)

PET values are lower than the monthly rainfall during the month of May to October indicating water surplus for possible recharge into groundwater regime during these months. The monthly PET ranges from 119.3 to 177.0mm.

CHAPTER 2

LITERATURE SURVEY

1. Zhaoshi Wu, Xijun Lai, Kuanyi Li (2021) "Water quality assessment of rivers in Lake Chaohu Basin (China) using water quality index": Water quality refers to the chemical, physical, and biological characteristics of water based on the standards of its usage. Water quality can be evaluated by environmental parameters using multiple methods, including statistical approach, modeling technique, water quality index (WQI), and so on. Among these methods, WQI plays a crucial role in water resource management, which is widely used in surface water (especially rivers) and groundwater. Various environmental parameters, such as dissolved oxygen (DO), metals, and nutrients, can be effectively converted into a WQI score representing the water quality status.

2. CPCB, Delhi: CPCB is monitoring the water quality in India under National Water Quality Monitoring Programme. Priority levels of polluted stretch is based on the risk. Risk is defined as; Risk= frequency of violation of criteria x consequence (magnitude). The degree of violation is with respect to water quality criteria for drinking water source with conventional treatment with respect to BOD (Annexure-I- Water Quality Criteria). The polluted locations in a continuous sequence are defined as polluted river stretches.

Criteria for Priority 1

• Monitoring locations exceeding BOD concentration 30 mg/l has been considered as it is the standard of sewage treatment plant and in river it appears without dilution. (River locations having water quality exceeding discharge standards for BOD to fresh water sources).

• All monitoring locations exceeding BOD concentration 6 mg/l on all occasions.

• Monitoring locations exceeding 3 mg/l BOD are not meeting desired water quality criteria but does not affect to Dissolved Oxygen level in water bodies. If BOD exceeds 6mg/l in water body, the Dissolved Oxygen is reduced below desired levels.

• The raw water having BOD levels up to 5 mg/l are does not form complex chemicals on chlorination for municipal water supplies. Hence the water bodies having BOD more than 6 mg/l are considered as polluted and identified for remedial action.

Criteria for Priority 2

• Monitoring locations having BOD between 20-30 mg/l.

• All monitoring locations exceeding BOD concentration 6 mg/l on all occasions.

Criteria for Priority 3

• Monitoring locations having BOD between 10-20 mg/l.

• All monitoring locations exceeding BOD concentration 6 mg/l on all occasions.

Criteria for Priority 4

• Monitoring locations having BOD between 6-10 mg/l.

Criteria for Priority 5

• Monitoring locations having BOD between 3-6 mg/l.

• The locations exceeding desired water quality of 3mg/l BOD.

3. For rivers, the primary sampling point is in the surface water layer (0-5 cm from the surface) at the centre of the main flow. However, the top 1-2 cm of this surface layer should be avoided so as not to collect floating dust, oil, etc. In addition, further samples can be collected through the full depth of the water column if required to meet the purpose of the study. The type of water sampling tool required will depend on the sampling site and the type of sample to be taken. The number and volume (size) of the samples to be collected will depend on the number and concentration of the target analytes, the difficulty and expense of analysis, and whether extra storage is required. Label each sample unambiguously, i.e., write on the sample bottle in water-resistant ink details of the name or code of the sample, the sampling date, the sampling site name etc.

4. CPCB, Bangalore: Grab samples are also called as spot or catch samples. Grab samples are single samples collected at a specific spot at a site in specified time. Grab samples are to be collected only when the source is known to be constant in composition for an extended period of time. Examples are, ground water samples, well mixed surface waters, large lakes, rivers, estuaries, shorelines, wastewater streams that are expected to be constant in composition over an extended period of time, like spent wash line in a distillery.

When the source composition varies from location to location, like upstream and downstream of a river, then grab samples can be collected from appropriate locations. This helps in finding out the extent of variation and duration of variation.

5. Clair N. Sawyer; Perry L. McCarty; Gene F. Parkin (2003):

Biochemical Oxygen Demand (BOD): When biodegradable organic matter (including organic waste) is present in waters it provides nutrient for the growth of bacteria and other microorganisms causing them to multiply and - where bacterial numbers are sufficient causing a depletion of dissolved oxygen in the water. The BOD (5day) test is a measure of the amount of oxygen consumed by microorganisms in breaking down the organic matter. The BOD value in rivers often increases during periods of heavy rain and high river flows as organic matter is washed in from the land.

Chloride: Chloride occurs naturally in river waters. Some effluents may contain elevated chloride levels and changes in concentrations may be an indicator of pollution. However very high levels of chloride may be seen especially in the lower reaches of rivers where a tidal influence exists. Its presence in lakes adjacent to coastal areas is generally due to saline deposition from rainfall and is correlated with Sodium concentrations.

Dissolved Oxygen (DO): Dissolved Oxygen is essential for the survival of fish and other aquatic life, and is an important indicator of pollution and/or eutrophication in rivers. The solubility of oxygen in river waters depends mainly on the water temperature and salinity. Water in equilibrium with air is normally saturated with oxygen (i.e., at 100% Saturation) and DO can be expressed either as % Saturation or in concentration terms as mg/l O2. Several factors can cause deviations in DO from equilibrium conditions and there can be a time lag before the equilibrium is restored. Such variations can be caused by: River morphology e.g., rapids or other areas of turbulence, Seasonal changes in temperature, Consumption of oxygen by microorganisms in the breakdown of organic matter present. This can lead to an oxygen sag where a sufficient amount of organic matter is present.

Eutrophication: The exchange of oxygen between plants and water in eutrophic waters can cause diurnal variations in DO leading to high (frequently super-saturated) levels of dissolved oxygen during the day but dissolved oxygen levels can decline seriously during the hours of darkness with the critical period for dissolved oxygen in eutrophic waters occurring just before dawn (i.e., after maximum hours of darkness).

pH: pH is a measure of the acidity of water and varies depending on the geology of the river catchment, on river flow, and on wastewater discharges but is generally in the range 6 – 9. It can also be influenced by biological processes, chiefly carbon dioxide uptake by plants during

photosynthesis. In eutrophic waters, diurnal variations of pH may follow the diurnal variations in dissolved oxygen.

Temperature: Water temperature is a major influence on the rate of biological growth with variations in Irish river waters being both seasonal and weather dependant. Localized elevated temperatures can occur where there are discharges such as industrial cooling waters.

CHAPTER 3

OBJECTIVES

- To study the physical and chemical characterstics of Mogral river water.
- To find the water quality index of river Mogral.
- To check whether the water quality of the river Mogral is in compliance with Bureau of Indian Standard (BIS).

CHAPTER 4

METHODOGY

4.1 SAMPLING LOCATION

The Mogral River is a west-flowing river in the Kasaragod district of Kerala. It gets its name from Mogral, the coastal village on its northern banks. The river empties into the Arabian Sea in MogralPuthur. It has a length of 34 kilometres and a drainage area of 132 (119.997) square kilometres.

The river originates in Kanathur in the Karadka Reserve Forest, in Kasaragod. Nettipadi and Muliyar are the tributaries feeding into the river. After flowing in a north-westerly direction, it is joined by another stream rising from the northern part of Karadka Reserve Forest. From Yedhir, the river takes a meandering course and flows through the fairly flat regions of Madhur and Patla. Toward the end of its course, it forms a long stretch of backwaters several kilometres long.

The river provide ample irrigation facilities. Mogral is a river of plains and the landuse is mainly dry land cultivation. The built-up are plenty along the southern edge and waste land along the northern edge. There are no tributaries and small streams from valleys on either side join the river. There is no forest inside the river basin. The Mogral Puthur and Madhur Panchayats have considerable amount of water bodies and paddy fields.

The river flows through paddy fields, plantations etc. The soil is gravelly laterite in midland and hilly areas where as alluvial and sandy in coastal plains. The annual average rainfall is about 3600 mm. The major source of rainfall is southwest monsoon from June to September which contributes nearly 85.3% of the total rainfall of the year. The northeast monsoon contributes nearly 8.9% and a balance of 5.8% is received during the month of January to May as pre monsoon showers.

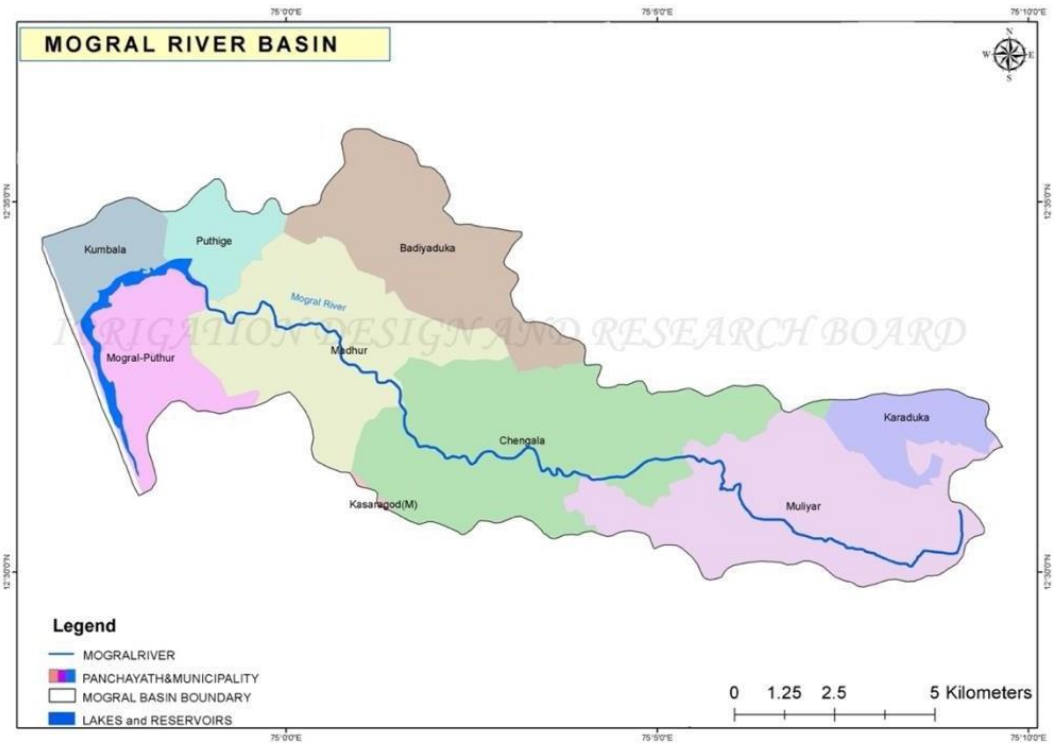

Fig no 4.1 Mogral river basin

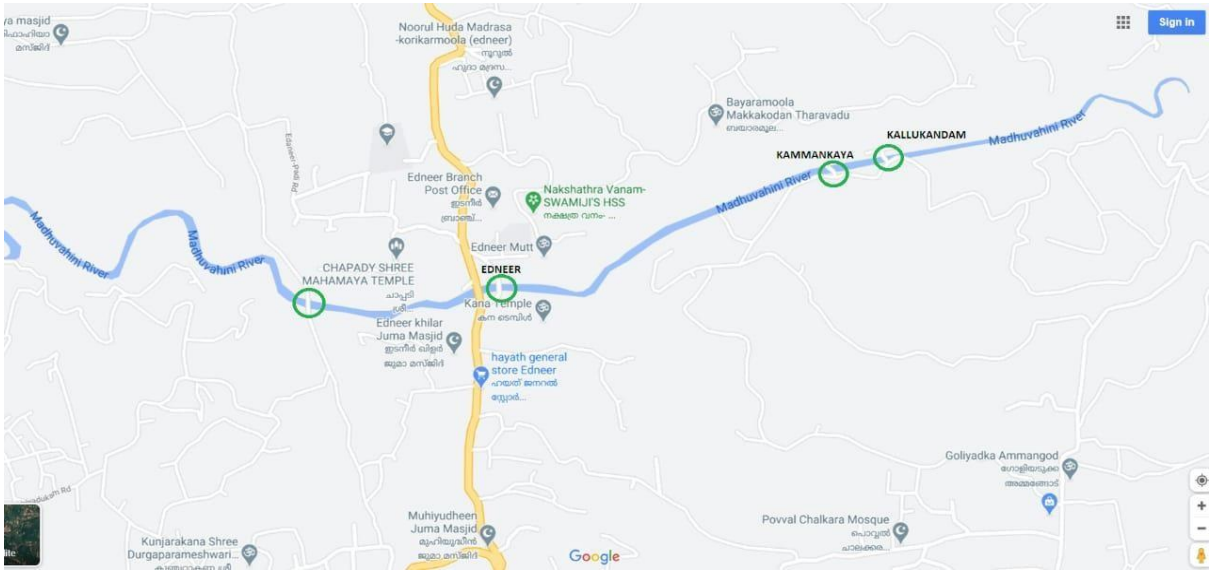

Fig no 4.2 Sampling locations: Kallukandam, Kammankaya, Edneer Madam, Edneerkunjar, Chappady.

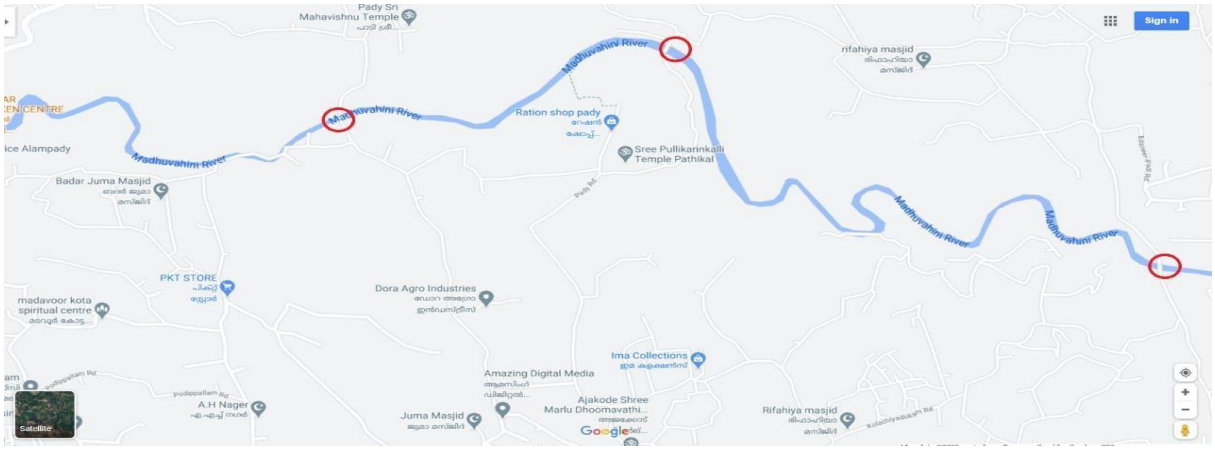

Fig no 4.3 Sampling locations: Pady bridge, Ethirthodu, Pady bellur.

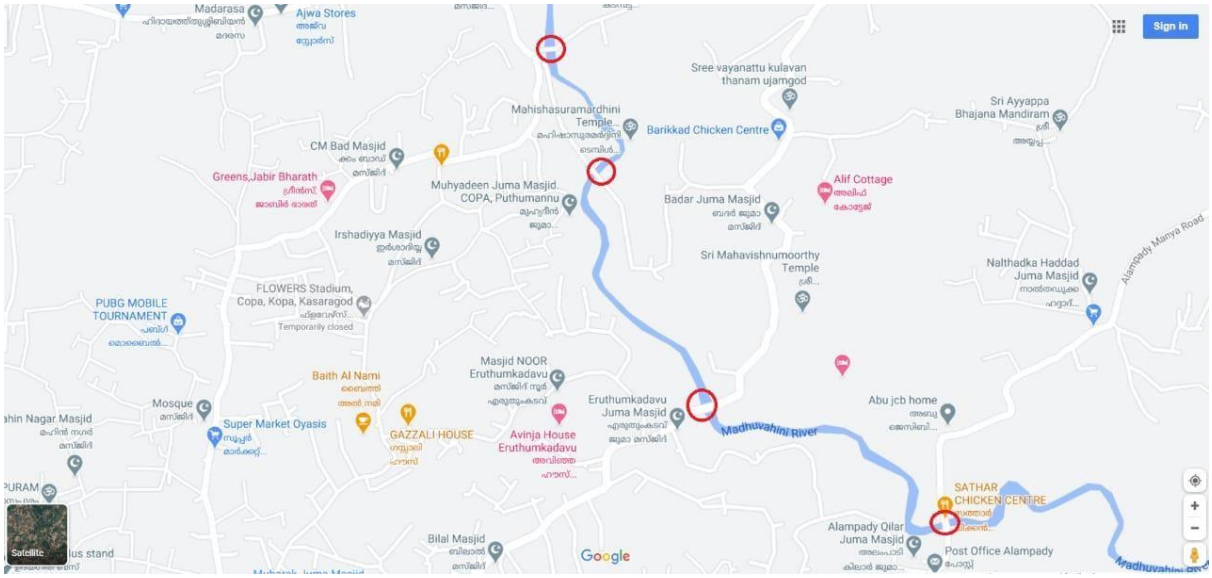

Fig no 4.4 Sampling locations: Alampady, Eruthumkadav, Barikad, Kopa.

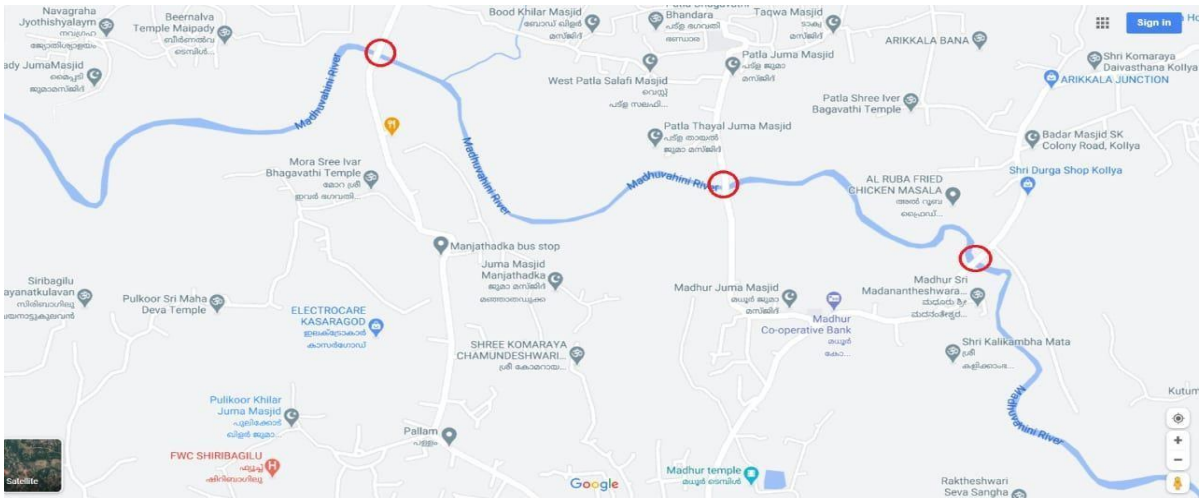

Fig no 4.5 Sampling locations: Madhur temple, Patla, Manjathadukka, Patla tributary.

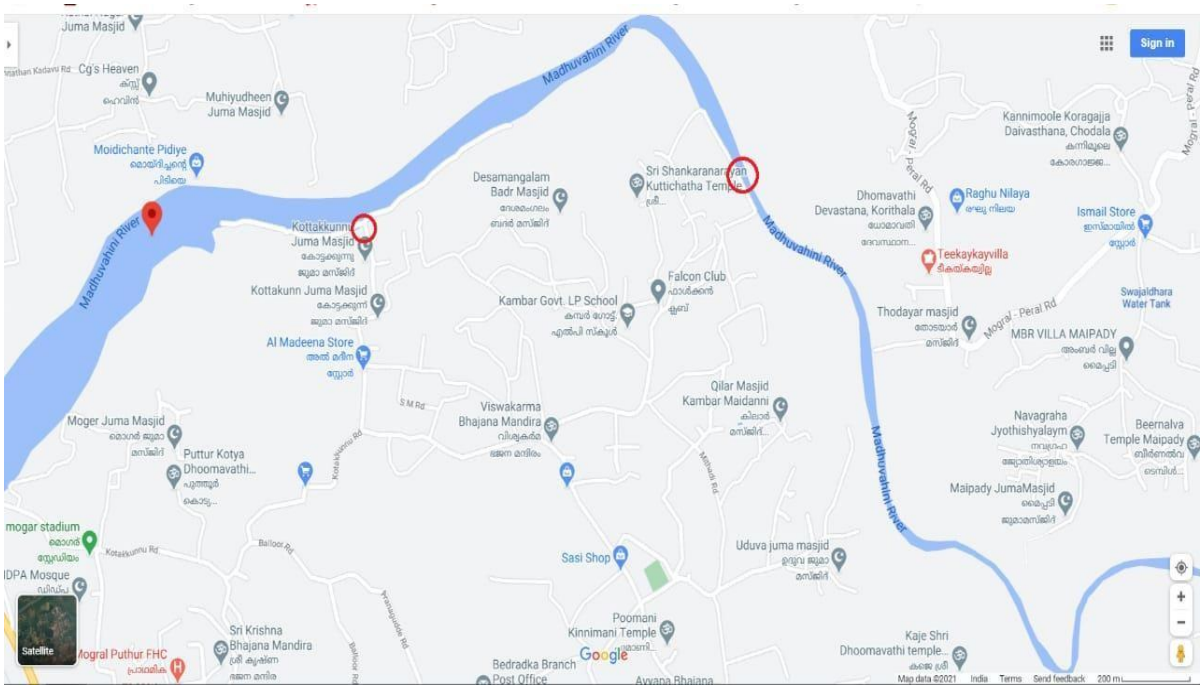

Fig no 4.6 Sampling locations: Maipapady, Kambar, Kotakkunnu.

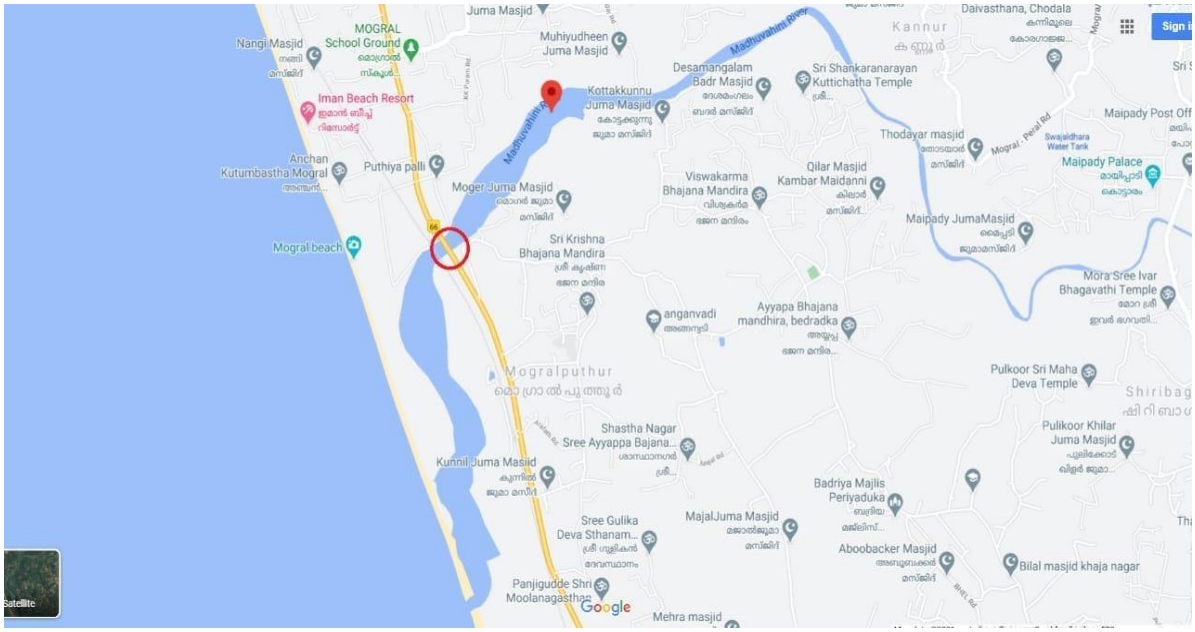

Fig no 4.7 Sampling locations: Mogral kadavath, Mogral bridge

4.2 SAMPLE COLLECTION AND ANALYSIS

The river water samples were collected from the selected twenty four locations (S1–S24) given in table as per standard sampling methods (IS: 2498, 1966 – Part-I) in the month of January and February. Analysis of various parameters namely temperature, pH, dissolved oxygen, Bio chemical Oxygen Demand, Alkalinity, Chloride were carried out as per the standard guidelines and procedures (APHA, 2012).). All the chemicals and reagents used were of analytical grade.

Figure no 4.8 Site visit and sampling

Table 4.1: Details of sampling stations.

STATION ID	SAMPLING STATIONS	LONGITUDE	LATITUDE
S1	Kallukandam	75.082	12.526
S2	Kammankaya	75.081	12.525
S3	Edaneer Madam	75.072	12.522
S4	Edneerkunjar	75.07	12.522
S5	Chappady	75.067	12.522
S6	Pady Bridge	75.054	12.529
S7	Ethirthodu	75.051	12.529
S8	Padybellur	75.045	12.527
S9	Alampady	75.035	12.526
S10	Eruthumkadav	75.028	12.529
S11	Barikkad	75.025	12.536
S12	Kopa	75.024	12.539
S13	Aranthodu	75.021	12.544
S14	Chenakkad	75.012	12.549
S15	Madhur Temple	75.01	12.554
S16	Patla	75.003	12.557
S17	Manjathadukka	74.998	12.556
S18	Patla tributary	74.995	12.56
S19	Maippady	74.993	12.561
S20	Kambar	74.977	12.568
S21	Kotakkunnu	74.967	12.566
S22	MogralKadavath	74.959	12.563
S23	Mogral Bridge	74.955	12.56
S24	Azad Nagar	74.957	12.549

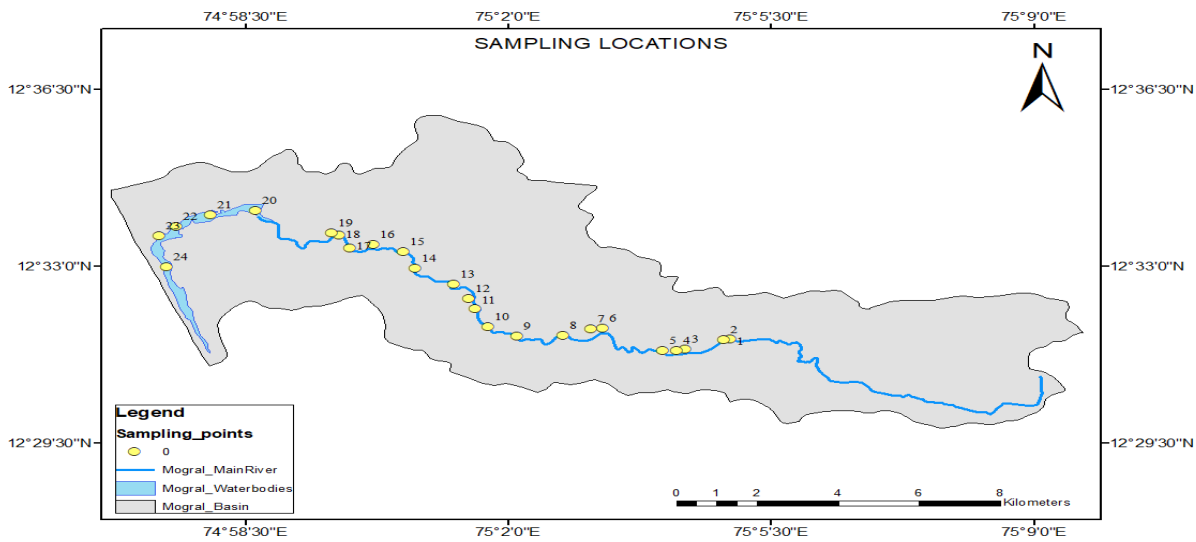

Fig no 4.9 Map showing water quality sampling stations

4.3 TESTING PARAMETERS

4.3.1 pH

According to the standard proposed by BIS (listed in Table 3.2), the range of Ph lies between 6.5 to 8.5. If the pH is less than 6.5, it discontinues the making of vitamins and minerals in the human body. More than 8.5 pH values cause the taste of water more salty and causes eye irritation and skin disorder for pH of more than 11. The rainwater which has no minerals useful for human body has a pH of 5.5–6 and not harmful on used as drinking purpose. pH in the range 3.5–4.5 affects the aquatic life.

4.3.2 Dissolved Oxygen

The dissolved oxygen reveals the changes occur in the biological parameters due to aerobic or anaerobic phenomenon and signifies the condition of the river/streams water for the purpose of the aquatic as well as human life. The aquatic life disturbed due to low values of DO. As per the standards proposed by BIS (Table 1), the range of DO lies between 4 to 6 mg/Lensures better aquatic life in the water body.

4.3.3 Biochemical Oxygen Demand

The standard given by BIS (Table 1) for BOD is 2–3 mg/L for class A, B, C respectively. BOD is used for determination of requirement of oxygen for stabilizing household and industrial wastes. The effluents disposed by domestic and industries into the surface and ground water contaminate the quality of the water which can be assessed by BOD determination.

4.3.4 Alkalinity

The alkalinity is a measure of the ability of water body to neutralize acids and bases and thus maintain a fairly stable pH level. Alkaline compounds in water such as bicarbonates, carbonates and hydroxides remove H+ ions and lower acidity. The standard given by BIS for Alkalinity is 200 mg/L.

4.3.5 Chloride

Chloride occurs in all natural waters in widely varying concentration. It is present in the water as salts of calcium, sodium and potassium. Chloride increases the electrical conductivity of water and thus increases its corrosivity.As the mineral content increases chloride content also increases. High chloride content indicates heavy pollution.The standard given by BIS for chloride is 250mg/L.

4.4 EXPERIMENTAL PROCEDURES

4.4.1 Determination of pH

APPARATUS

1. pH meter with electrodes
2. Thermometer
3. pH papers

REAGENTS

1. Buffer solutions of known pH values.

PROCEDURE

1. Rinse the electrode with D.W and dry using a tissue paper.
2. Immerse the electrode in a suitable buffer solution.
3. Make the necessary corrections for temperature.
4. Measure the pH and adjust the calibration knob to bring the reading correct value.
5. Rinse the electrode with D.W and dry it using a tissue paper. Immerse the electrode in a buffer solution at the other extreme of the range, and adjust the value using correction knob.
6. Rinse the electrode with D.W and dry it using a tissue paper. Place the electrode in the sample whose pH is to be measured and note the reading.
7. After measurement dip and store the electrode in 4M KCl solution.

4.4.2 Determination of Dissolved Oxygen (DO)

APPARATUS

1. Burette
2. Pipette
3. 300 ml glass stoppered BOD bottles
4. Conical flask
5. Measuring cylinder

REAGENTS

1. Manganous sulphate solution ($MnSO_4.4H_2O$)
2. Alkali-iodide-Azide reagent
3. Conc.sulphuric acid(36N)
4. Starch indicator
5. Standard sodium thiosulphate solution(0.025N)

PROCEDURE

1. Take two 300ml glass stoppered BOD bottle and fill it with sample to be tested. Avoid any kind of bubbling and trapping of air bubbles.
2. Add 2 ml of manganous sulphate to the BOD bottle by inserting the calibrated pipette just below the surface of the liquid.
3. Add 2 ml of alkali-iodide-Azide reagent in the same manner. If oxygen is present, a brownish-orange cloud of precipitate or flock will appear.
4. Insert stopper with care to exclude air bubbles and mix by inverting the bottle atleast 15 times.
5. When the precipitate settles, leaving a clear supernatant above the manganese hydroxide floc, shake again.
6. After 2 minutes of settling, carefully remove the stopper, immediately add 2ml concentrated sulphuric acid by allowing the acid to run down the neck of the bottle.
7. Insert stopper and mix by gentle inversion until dissolution is complete.
8. Rinse the burette with sodium thiosulphate and then fill it with sodium thiosulphate. Fix the burette to the stand.
9. Measure out 203 ml of the solution from the bottle to an Erlenmeyer flask. As 2 ml each of manganese sulphate and azide reagent have been added, the proportionate quantity of yellow solution corresponds to 200 ml of sample is 203 ml.
10. Titrate with 0.025N sodium thiosulphate solution to a pale straw colour.
11. Add 1–2 mL starch solution and continue the titration to the first disappearance of the blue colour and note down the volume of sodium thiosulphate solution added, which gives directly the D.O. in mg/l.

4.4.3 Determination of Biological Oxygen Demand (BOD)

APPARATUS

1. B.O.D.bottles 300 mL capacity
2. B.O.D.incubator
3. Burette
4. Pipette
5. Conical flask
6. Measuring cylinder

REAGENTS

1. Manganous sulphate solution (MnSO4.4H2O)
2. Alkali-iodide-Azidereagent
3. Conc.sulphuric acid (36N)
4. Starch indicator
5. Standard sodium thiosulphate solution(0.025N)

PROCEDURE

1. Take two 300ml glass stoppered BOD bottle and fill it with sample to be tested. Avoid any kind of bubbling and trapping of air bubbles.
2. Add ml of sample to each bottle and fill the remaining quantity with the dilution water.
3. Preserve one bottle in a BOD incubator at 20°C for five days.The other one bottle needs to be analysed immediately to find out initial D.O.
4. Add 2ml of manganous sulphate to that BOD bottle by inserting the calibrated pipette just below the surface of the liquid.
5. Add 2ml of alkali-iodide-Azide reagent in the same manner. If oxygen is present, a brownish-orange cloud of precipitate or flock will appear.
6. Stopper with care to exclude air bubbles and mix by inverting the bottle atleast 15 times.
7. When the precipitate settles, leaving a clear supernatant above the manganese hydroxide floc, shake again.
8. After 2 minutes of settling, carefully remove the stopper, immediately add 2ml concentrated sulphuric acid by allowing the acid to run down the neck of the bottle.
9. Insert stopper and mix by gentle inversion until dissolution is complete.

10. Rinse the burette with sodium thiosulphate and then fill it with sodium thiosulphate. Fix the burette to the stand.
11. Measure out 203ml of the solution from the bottle to an Erlenmeyer flask.
12. Titrate with 0.025N sodium thiosulphate solution to a pale straw colour.
13. Add 1–2 mL starch solution and continue the titration to the first disappearance of the blue colour.
14. Note down the volume of sodium thiosulphate solution added, which gives directly the initial D.O. in mg/l.
15. After three days, take out the bottle from the BOD incubator and analyse the sample for final D.O.
16. Repeat the steps 4 to 13 to find out the final D.O. Note down the volume of sodium thiosulphate solution added to the second bottle, which gives directly the final D.O. in mg/l.

4.4.4 Determination of alkalinity

APPARATUS

1. Burette
2. Pipette
3. Erlenmeyer flask
4. Measuring cylinder
5. Wash bottle and beakers

REAGENTS

1. 0.02N sulphuric acid
2. Phenolphthalein indicator
3. Methyl orange indicator

PROCEDURE

1. Rinse the burette with 0.02N Sulphuric acid and discard thesolution.
2. Fill the burette with 0.02N sulphuric acid and adjust it to zero.
3. Fix the burette in the stand. Using a measuring cylinder exactly measure 50mL of sample and pour it into a 250 mL of conical flask.
4. Add few drops of phenolphthalein indicator to the contents of conical flask. The colour of the solution will turn to pink (at pH 8.3). This colour change is

due to alkalinity of hydroxylions in the water sample.

5. Titrate it against 0.02N sulphuric acid till the pink colour disappears. This indicates that all the hydroxylions are removed from the water sample.
6. Note down the titter value. This value is used for calculating the phenolphthale in alkalinity.
7. Add a few drops of methyl orange to the same sample.Colour turns yellow.
8. Continue titration against 0.02N sulphuric acid. When the pH value reaches 4.5 the sample
9. Becomes orange yellow in colour.
10. Note the amount of acid consumed from the beginning and calculate the total alkalinity.
11. Express all the types of alkalinity.

4.4.5 Determination of chloride

APPARATUS

1. Burette
2. Pipette
3. Ernenmeyer flask
4. Measuring cylinder

REAGENTS

1. Chloride free distilled water.
2. Standard silver nitrate solution(0.0141N).
3. Potassium chromate indicator.
4. Acid or alkali for adjusting pH.

PROCEDURE

A. Sample titration

1. Before starting the titration rinse the burette with silver nitrate solution. Fill the burette with silver nitrate solution. Adjust to zero and fix the burette in stand.
2. Take 50 ml of the sample in a clean 250mL conical flask and dilute it to 100ml.
3. Sample is brought to pH 7-8 by adding acid or alkali as required.

4. Add 1ml of Potassium Chromate indicator to get light yellow colour
5. Titrate the sample against silver nitrate solution until the colour changes from yellow to brick red.i.e., the end point.
6. Note the volume of Silver nitrate added.

B. Blanktitration

1. Take 50ml of the distilled water in a clean 250 ml conical flask.
2. Add 1ml of Potassium Chromate indicator to get light yellow colour.
3. Titrate the sample against silver nitrate solution until the colour changes from yellow to brick red. i.e., the end point.
4. Note the volume of silver nitrate added for distilled water.

4.5 WATER QUALITY STANDARDS

Water Quality Standards are provisions of state, territorial, authorized tribal or federal law approved by EPA that describe the desired condition of a water body or the level of protection or mandate how the desired condition will be expressed or established for such waters in the future.

The water quality have been assessed against the standards developed by Bureau of Indian Standards (BIS).

Surface Water Quality Standards (as per IS: 2296).

Class A – Drinking water without conventional treatment but after disinfection. Class B – Water for outdoor bathing. Class C – Drinking water with conventional treatment followed by disinfection. Class D – Water for fish culture and wild life propagation. Class E – Water for irrigation, industrial cooling and controlled waste disposal. (Unobj = Unobjectionable).

Table 4.2: Surface water quality standards (as per IS: 2296).

SL	Parameter and Unit	A	B	C	D	E
1	pH (max) (min : 6.5)	8.5	8.5	8.5	8.5	8.5
2	DO (mg/L) (minimum)	6	5	4	4	--
3	BOD (3d, 27oC) (mg/L)	2	3	3	--	--
4	Chlorides (mg/L as Cl)	250	--	600	--	600

Table 4.3: Drinking Water Quality Standards (IS: 10500)

SL	Parameters and U nits	Standards
1	Alkalinity (mg/L as $CaCO_3$)	200

4.6 WATER QUALITY INDEX

WQI indicates the quality of water in terms of index number which represents overall quality of water for any intended use. It is defined as a rating reflecting the composite influence of different water quality parameters were taken into consideration for the calculation of water Quality index (WQI) .The indices are among the most effective ways to communicate the information on water quality trends to the general public or to the policy makers and in water quality management. In formulation of water quality index the relative importance of various parameters depends on intended use of water. Mostly it is done from the point of view of its suitability for human consumption.

4.6.1 Determination of water quality index

Weighted Arithmetic WQI proposed by Horton (1965) and used five parameters pH (units), Dissolved Oxygen (mg L−1), Biological Oxygen Demand (mg L−1), Alkalinity , Chloride and Temperature (◦C) for the calculation of WQI given by the following equations:

Step 1: In the first step, unit weight (Wi) for various parameters is inversely proportional to the recommended standard (Sstandard) for the corresponding parameter. Wi values were calculated by using the following formula proposed by Tiwari and Mishra (1985),

$$Wi = K\sum \frac{1}{Sstandard}$$

The constant of proportionality K in the above equation can be determined from the following equation,

$$K = \frac{1}{\sum \frac{1}{S1} + \frac{1}{S2} + \cdots + \frac{1}{Sn}}$$

Step 2: Calculate quality rating scale (Qi) of *i*th parameter for a total of *n* water quality parameters is calculated by using this equation,

$$Qi = \left(\frac{Qactual - Qideal}{Sstandard - Qideal}\right) \times 100$$

Step 3: Finally, the overall WQI was calculated by aggregating the quality rating with the unit weight linearly using the following equation:

$$WAWQI = \sum_{i=1}^{i=n} \frac{QiWi}{Wi}$$

In this study, the WQI was considered for human consumption or uses, and the maximum permissible WQI for the drinking water was taken as 100 score. Rating scale proposed was in the range of 0–100 and grading were proposed as A for 0–25 (Excellent), B for 26–50 (Good), C for 51–75 (Poor), D for 76–100 (Very Poor) and above 100 (Unsuitable for drinking purpose) stands for E.

Where

Wi= unit weight for each water quality parameter

K = proportionality constant

Qi= the quality rating scale for each parameter

Qactual= estimated concentration of ith parameter in the analyzed water

Qideal= the ideal value of this parameter in pure water, Qideal= 0 (except pH = 7.0 and DO = 14.6 mg L^{-1})

Sstandard= recommended standard value of ith parameter

n = number of water quality parameters.

Table 4.4: Water quality rating as per weighted arithmetic water quality index method

WQI Value	**Rating of Water Quality**	**Grading**
0-25	Excellent Water Quality	A
26-50	Good Water Quality	B
51-75	Poor Water Quality	C
76-100	Very Poor Water Quality	D
Above 100	Unsuitable for drinking purpose	E

CHAPTER 5

RESULTS AND DISCUSSIONS

This study shows variations for the different parameters in the month of January and February at various stations (S1–S24) with their ranges have been tabulated and shown in table 5.1 and 5.2

Temperature of water is basically important because it effects bio-chemical reactions in aquatic organisms. A rise in temperature of water leads to the speeding up of chemical reactions in water, reduces the solubility of gases and amplifies the tastes and odours. The average temperature of the present study ranged from 28℃ to 30℃.

From Table 4.1 and 4.2, it is clear that the range of the pH lies between 6.1 to 7.26. Majority of the sites followed the standards given by BIS shown in Table 1. S24 have value less than prescribed value in January and S07 and S18 in both months. In the present study a narrow variation of pH is observed for all sampling stations (S1–S24) is due to low annual variation in free CO2.

The DO values obtained in the present study are between 5.8mg/L to 7.9mg/L, which is in the standard range given by BIS. S18 recorded as the lowest DO value among all the sites.

In the present investigation the total alkalinity of the water samples is found in the Range 10mg/L to 130 mg/L, which is in the prescribed limit.

Chloride values obtained in the study are found in the range between 18mg/L to 18899.47mg/L, which is much greater than the standard value. In January, most of sites showed higher chloride value while in February, it was reduced and only 6 sites resulted higher value than prescribed value. The high concentration of chloride is considered to be an indication of pollution due to sea water intrusion and lead to water quality decline so that the corresponding water becomes inadequate for specific purposes including drinking, agricultural, and industrial uses.

BOD values falls between 0 to 2.1mg/L for the selected sites. It is observed that maximum value of BOD is found at site S18.

Table 5.1: physico-chemical water quality parameters of Mogral river in January 2021

JANUARY 2021						
Station ID	**Temperature ^{0} (C)**	**pH**	**DO (mg/L)**	**BOD (mg/L)**	**Chlorides (mg/L)**	**Alkalinity (mg/L)**
S01	29	6.52	6.4	0.3	62	54
S02	29.5	6.71	6.6	0.4	87	52
S03	28	6.9	6.8	1.1	264	40
S04	28	7	7.3	1.4	276	64
S05	29	6.81	7.6	1.2	270	80
S06	30	6.42	6.5	1.1	276	114
S07	33	6.3	7.1	0.5	287	40
S08	29.5	7.1	7.5	0.2	299	25
S09	28.5	6.63	7.4	0.7	291	20
S10	30	6.82	6.7	0.6	299	114
S11	29.5	6.67	7.2	0.2	88	60
S12	29	6.63	7.4	0.3	276	54
S13	29	6.81	7.6	0.9	381	14
S14	29	7.1	6.9	1.4	241	58
S15	30	6.51	6.9	0.4	210	10
S16	30	6.76	7.1	0.4	428	32
S17	29	6.67	7.2	0.8	18	40
S18	29	6.2	5.9	2	264	20
S19	29	6.84	6.9	0.9	165	30
S20	29	6.76	7.6	0.3	920	80
S21	29	6.62	7.2	0.7	1300	100
S22	29	6.51	7	0.7	11100	40
S23	29	6.54	6.9	1.4	14300	54
S24	30	6.35	6.4	1.6	13899	86

Table 5.2: physico-chemical water quality parameters of Mogral river in February 2021

FEBRUARY 2021						
Station ID	**Temperature (^{0}C)**	**pH**	**DO (mg/L)**	**BOD (mg/L)**	**Chlorides (mg/L)**	**Alkalinity (mg/L)**
S01	29	6.58	6.6	0.2	72.50	40
S02	30	7.1	6.6	0.2	77.50	40
S03	30	6.9	6.9	0.7	112.50	25
S04	30	6.9	7.6	0	152.50	25
S05	29	6.72	7.9	1.4	122.50	25
S06	30	6.67	6.8	1	117.50	20
S07	30	6.4	7.2	0.5	117.50	25
S08	30	6.96	7.3	0.2	112.50	25
S09	29	7.1	7.9	1.2	122.50	20
S10	29	6.97	6.8	0.3	112.50	25
S11	29	6.72	7.2	0.2	127.50	25
S12	29	6.63	7.4	0.3	122.50	25
S13	29	6.81	7.6	0	742.48	35
S14	29	7.26	7	1.4	122.50	30
S15	30	6.51	6.9	0.3	122.50	30
S16	30	6.84	7.1	0.1	112.50	25
S17	29	6.56	7.3	0.9	122.50	25
S18	29	6.1	5.8	2.1	112.50	30
S19	29	6.76	6.8	0.7	142.50	30
S20	29	6.8	7.8	0.1	449.99	45
S21	29	6.89	7.1	0.8	1099.97	115
S22	29	6.54	7	0.7	14099.61	130
S23	29	6.67	7.2	1.3	16299.54	125
S24	30	6.71	6.5	1.6	18899.47	125

5.1 WATER QUALITY INDEX

WQI for six analyzed physico-chemical parameters of river Mogral at sites S1–S25 was calculated using weighted arithmetic method as per the procedure described earlier. The water quality index values are shown in table 5.6.

The values for water quality index ranges from good to very poor in the January and good to unfit in the month of February. In the month of January, water quality was poor for S6 and S22 and very poor for S23 and S24. Water quality of other sites were good.

In the month of February, S6 turned good while S14 and S18 become poor and S22 very poor. While S23 remained very poor, S24 become unfit.

Table 5.3: BIS standard, unit weight and ideal value of each parameter

Parameters	BIS Standards (Sn)	1/Sn	∑1/Sn	K=1/(∑1/Sn)	Wi=K/Sn	Ideal value (V)
pH	8.5	0.117647	0.6599804	1.5151965	0.178258415	7
DO	5	0.2	0.6599804	1.5151965	0.303039306	14.6
BOD	3	0.333333	0.6599804	1.5151965	0.50506551	0
Chlorides	250	0.004	0.6599804	1.5151965	0.006060786	0
Alkalinity	200	0.005	0.6599804	1.5151965	0.007575983	0
		0.65998			1	

Table 5.4: Calculation of water quality of sampling stations S1-S24 (January 2021)

SITE NO	Mean conc. value Vn	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 1	6.54	0.3066667	30.66666667	5.4665914	35.58548283
	6.5	0.84375	84.375	25.5689414	
	0.25	0.0833333	8.333333333	4.20887925	
	67.25	0.269	26.9	0.16303515	
	47	0.235	23.5	0.17803559	
				35.5854828	
SITE 2					
	6.9	0.0666667	6.66666667	1.18838944	31.8659675
	6.6	0.8333333	83.33333333	25.2532755	
	0.3	0.1	10	5.0506551	
	82.25	0.329	32.9	0.19939986	
	46	0.23	23	0.1742476	
				31.8659675	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 3	6.9	0.0666667	6.66666667	1.18838944	41.38395229
	6.85	0.8072917	80.72916667	24.4641106	
	0.9	0.3	30	15.1519653	
	188.25	0.753	75.3	0.45637719	
	32.5	0.1625	16.25	0.12310972	
				41.3839523	
SITE 4	6.95	0.0333333	3.33333333	0.59419472	35.63714658
	7.45	0.7447917	74.47916667	22.570115	
	0.7	0.2333333	23.33333333	11.7848619	
	214.25	0.857	85.7	0.51940937	
	44.5	0.2225	22.25	0.16856561	
				35.6371466	
SITE 5	6.76	0.16	16	2.85213464	47.03606514
	7.75	0.7135417	71.35416667	21.6231171	
	1.3	0.4333333	43.33333333	21.8861721	
	196.25	0.785	78.5	0.47577171	
	52.5	0.2625	26.25	0.19886954	
				47.0360651	
SITE 6					
	6.535	0.31	31	5.52601087	49.02952554
	6.65	0.828125	82.8125	25.0954425	
	1.05	0.35	35	17.6772928	
	196.75	0.787	78.7	0.47698387	
	67	0.335	33.5	0.25379542	
				49.0295255	
SITE 7					
	6.35	0.4333333	43.33333333	7.72453133	40.27282995
	7.15	0.7760417	77.60416667	23.5171128	
	0.5	0.1666667	16.66666667	8.4177585	
	202.25	0.809	80.9	0.4903176	
	32.5	0.1625	16.25	0.12310972	
				40.27283	
SITE 8					
	7.03	0.02	2	0.35651683	27.04507066
	7.4	0.75	75	22.7279479	
	0.2	0.0666667	6.666666667	3.3671034	
	205.75	0.823	82.3	0.4988027	
	25	0.125	12.5	0.09469978	
				27.0450707	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 9	6.85	0.1	10	1.78258415	40.29209523
	7.65	0.7239583	72.39583333	21.9387831	
	0.95	0.3166667	31.66666667	15.9937411	
	206.75	0.827	82.7	0.50122701	
	20	0.1	10	0.07575983	
				40.2920952	
SITE 10					
	6.885	0.0766667	7.666666667	1.36664785	34.48447518
	6.75	0.8177083	81.77083333	24.7797766	
	0.45	0.15	15	7.57598265	
	205.75	0.823	82.3	0.4988027	
	69.5	0.3475	34.75	0.2632654	
				34.4844752	
SITE 11					
	6.66	0.2266667	22.66666667	4.04052408	31.18911683
	7.2	0.7708333	77.08333333	23.3592798	
	0.2	0.0666667	6.666666667	3.3671034	
	107.75	0.431	43.1	0.26121988	
	42.5	0.2125	21.25	0.16098963	
				31.1891168	
SITE 12					
	6.615	0.2566667	25.66666667	4.57529933	32.98657269
	7.4	0.75	75	22.7279479	
	0.3	0.1	10	5.0506551	
	199.25	0.797	79.7	0.48304465	
	39.5	0.1975	19.75	0.14962566	
				32.9865727	
SITE 13	6.805	0.13	13	2.3173594	33.4445983
	7.6	0.7291667	72.91666667	22.0966161	
	0.45	0.15	15	7.57598265	
	561.74	2.24696	224.696	1.3618344	
	24.5	0.1225	12.25	0.09280579	
				33.4445983	
SITE 14	7.18	0.12	12	2.13910098	50.46456025
	6.95	0.796875	79.6875	24.1484447	
	1.4	0.4666667	46.66666667	23.5697238	
	181.75	0.727	72.7	0.44061915	
	44	0.22	22	0.16667162	
				50.4645602	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 15	6.505	0.33	33	5.8825277	36.56003842
	6.9	0.8020833	80.20833333	24.3062777	
	0.35	0.1166667	11.66666667	5.89243095	
	166.25	0.665	66.5	0.40304228	
	20	0.1	10	0.07575983	
				36.5600384	
SITE 16					
	6.77	0.1533333	15.33333333	2.7332957	31.38024946
	7.1	0.78125	78.125	23.6749458	
	0.25	0.0833333	8.333333333	4.20887925	
	270.25	1.081	108.1	0.65517098	
	28.5	0.1425	14.25	0.10795775	
				31.3802495	
SITE 17					
	6.58	0.28	28	4.99123563	42.79628975
	7.25	0.765625	76.5625	23.2014469	
	0.85	0.2833333	28.33333333	14.3101894	
	70.25	0.281	28.1	0.17030809	
	32.5	0.1625	16.25	0.12310972	
				42.7962897	
SITE 18					
	6.15	0.5666667	56.66666667	10.1013102	72.7859671
	5.85	0.9114583	91.14583333	27.6207701	
	2.05	0.6833333	68.33333333	34.5128098	
	188.25	0.753	75.3	0.45637719	
	25	0.125	12.5	0.09469978	
				72.7859671	
SITE 19					
	6.78	0.1466667	14.66666667	2.61445676	41.03335908
	6.85	0.8072917	80.72916667	24.4641106	
	0.8	0.2666667	26.66666667	13.4684136	
	153.75	0.615	61.5	0.37273835	
	30	0.15	15	0.11363974	
				41.0333591	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 20	6.75	0.1666667	16.66666667	2.97097359	30.01641984
	7.7	0.71875	71.875	21.7809501	
	0.2	0.0666667	6.666666667	3.3671034	
	684.995	2.73998	273.998	1.66064328	
	62.5	0.3125	31.25	0.23674946	
				30.0164198	
SITE 21					
	6.745	0.17	17	3.03039306	42.49049366
	7.15	0.7760417	77.60416667	23.5171128	
	0.75	0.25	25	12.6266377	
	1199.985	4.79994	479.994	2.90914097	
	107.5	0.5375	53.75	0.40720907	
				42.4904937	
SITE 22					
	6.52	0.32	32	5.70426929	72.34761148
	7	0.7916667	79.16666667	23.9906117	
	0.7	0.2333333	23.33333333	11.7848619	
	12599.805	50.39922	5039.922	30.5458893	
	85	0.425	42.5	0.32197926	
				72.3476115	
SITE 23					
	6.585	0.2766667	27.66666667	4.93181616	88.92302154
	7.05	0.7864583	78.64583333	23.8327788	
	1.35	0.45	45	22.7279479	
	15299.77	61.19908	6119.908	37.0914535	
	89.5	0.4475	44.75	0.33902522	
				88.9230215	
SITE 24					
	6.505	0.33	33	5.8825277	98.70266474
	6.45	0.8489583	84.89583333	25.7267744	
	1.6	0.5333333	53.33333333	26.9368272	
	16399.235	65.59694	6559.694	39.7569023	
	105.5	0.5275	52.75	0.39963308	
				98.7026647	

Table 5.5: Calculation of water quality index of sampling stations S1-S24 (February)

SITE NO	Mean conc. value Vn	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 1	6.58	0.28	28	4.991235628	33.93889698
	6.6	0.833333333	83.33333333	25.2532755	
	0.2	0.066666667	6.666666667	3.3671034	
	72.5	0.29	29	0.175762797	
	40	0.2	20	0.151519653	
				33.93889698	
SITE 2	7.1	0.066666667	6.666666667	1.188389435	30.14817236
	6.6	0.833333333	83.33333333	25.2532755	
	0.2	0.066666667	6.666666667	3.3671034	
	77.5	0.31	31	0.18788437	
	40	0.2	20	0.151519653	
				30.14817236	
SITE 3	6.9	0.066666667	6.666666667	1.188389435	37.64696416
	6.9	0.802083333	80.20833333	24.30627767	
	0.7	0.233333333	23.33333333	11.7848619	
	112.5	0.45	45	0.272735375	
	25	0.125	12.5	0.094699783	
				37.64696416	
SITE 4	6.9	0.066666667	6.666666667	1.188389435	23.74941323
	7.6	0.729166667	72.91666667	22.09661606	
	0	0	0	0	
	152.5	0.61	61	0.369707953	
	25	0.125	12.5	0.094699783	
				23.74941323	
SITE 5	6.72	0.186666667	18.66666667	3.327490419	48.43851075
	7.9	0.697916667	69.79166667	21.14961823	
	1.4	0.466666667	46.66666667	23.5697238	
	122.5	0.49	49	0.29697852	
	25	0.125	12.5	0.094699783	
				48.43851075	
SITE 6	6.67	0.22	22	3.921685136	45.73976252
	6.8	0.8125	81.25	24.62194361	
	1	0.333333333	33.33333333	16.835517	
	117.5	0.47	47	0.284856948	
	20	0.1	10	0.075759826	
				45.73976252	
SITE 7	6.4	0.4	40	7.130336611	39.28693168
	7.2	0.770833333	77.08333333	23.35927984	
	0.5	0.166666667	16.66666667	8.417758499	
	117.5	0.47	47	0.284856948	
	25	0.125	12.5	0.094699783	
				39.28693168	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 8	6.96	0.026666667	2.666666667	0.475355774	27.25350822
	7.3	0.760416667	76.04166667	23.04361389	
	0.2	0.066666667	6.666666667	3.3671034	
	112.5	0.45	45	0.272735375	
	25	0.125	12.5	0.094699783	
				27.25350822	
SITE 9	7.1	0.066666667	6.666666667	1.188389435	42.91336641
	7.9	0.697916667	69.79166667	21.14961823	
	1.2	0.4	40	20.2026204	
	122.5	0.49	49	0.29697852	
	20	0.1	10	0.075759826	
				42.91336641	
SITE 10	6.97	0.02	2	0.356516831	30.3965507
	6.8	0.8125	81.25	24.62194361	
	0.3	0.1	10	5.0506551	
	112.5	0.45	45	0.272735375	
	25	0.125	12.5	0.094699783	
				30.3965507	
SITE 11	6.72	0.186666667	18.66666667	3.327490419	30.45767353
	7.2	0.770833333	77.08333333	23.35927984	
	0.2	0.066666667	6.666666667	3.3671034	
	127.5	0.51	51	0.309100092	
	25	0.125	12.5	0.094699783	
				30.45767353	
SITE 12	6.63	0.246666667	24.66666667	4.39704091	32.56732226
	7.4	0.75	75	22.72794795	
	0.3	0.1	10	5.0506551	
	122.5	0.49	49	0.29697852	
	25	0.125	12.5	0.094699783	
				32.56732226	
SITE 13	6.81	0.126666667	12.66666667	2.257939927	26.28714068
	7.6	0.729166667	72.91666667	22.09661606	
	0	0	0	0	
	742.48	2.96992	296.992	1.800004991	
	35	0.175	17.5	0.132579696	
				26.28714068	
SITE 14	7.26	0.173333333	17.33333333	3.089812532	51.06076631
	7	0.791666667	79.16666667	23.99061172	
	1.4	0.466666667	46.66666667	23.5697238	
	122.5	0.49	49	0.29697852	
	30	0.15	15	0.11363974	
				51.06076631	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 15	6.51	0.326666667	32.66666667	5.823108233	35.59065926
	6.9	0.802083333	80.20833333	24.30627767	
	0.3	0.1	10	5.0506551	
	122.5	0.49	49	0.29697852	
	30	0.15	15	0.11363974	
				35.59065926	
SITE 16	6.84	0.106666667	10.66666667	1.901423096	27.62735573
	7.1	0.78125	78.125	23.67494578	
	0.1	0.033333333	3.333333333	1.6835517	
	112.5	0.45	45	0.272735375	
	25	0.125	12.5	0.094699783	
				27.62735573	
SITE 17	6.56	0.293333333	29.33333333	5.228913515	43.81617101
	7.3	0.760416667	76.04166667	23.04361389	
	0.9	0.3	30	15.1519653	
	122.5	0.49	49	0.29697852	
	25	0.125	12.5	0.094699783	
				43.81617101	
SITE 18	6.1	0.6	60	10.69550492	74.21506878
	5.8	0.916666667	91.66666667	27.77860305	
	2.1	0.7	70	35.3545857	
	112.5	0.45	45	0.272735375	
	30	0.15	15	0.11363974	
				74.21506878	
SITE 19	6.76	0.16	16	2.852134645	39.7180447
	6.8	0.8125	81.25	24.62194361	
	0.7	0.233333333	23.33333333	11.7848619	
	142.5	0.57	57	0.345464809	
	30	0.15	15	0.11363974	
				39.7180447	
SITE 20	6.8	0.133333333	13.33333333	2.37677887	26.78699161
	7.8	0.708333333	70.83333333	21.46528417	
	0.1	0.033333333	3.333333333	1.6835517	
	449.99	1.79996	179.996	1.090917258	
	45	0.225	22.5	0.17045961	
				26.78699161	
SITE 21	6.89	0.073333333	7.333333333	1.307228379	41.55287992
	7.1	0.78125	78.125	23.67494578	
	0.8	0.266666667	26.66666667	13.4684136	
	1099.97	4.39988	439.988	2.666673163	
	115	0.575	57.5	0.435619002	
				41.55287992	

SITE NO	Mean conc. value (Vn)	Vn/Sn	Vn/Sn*100=Qn	WnQn	Overall WQI
SITE 22	6.54	0.306666667	30.66666667	5.466591402	75.91639213
	7	0.791666667	79.16666667	23.99061172	
	0.7	0.233333333	23.33333333	11.7848619	
	14099.61	56.39844	5639.844	34.18188823	
	130	0.65	65	0.492438872	
				75.91639213	
SITE 23	6.67	0.22	22	3.921685136	89.1558463
	7.2	0.770833333	77.08333333	23.35927984	
	1.3	0.433333333	43.33333333	21.8861721	
	16299.54	65.19816	6519.816	39.51521032	
	125	0.625	62.5	0.473498916	
				89.1558463	
SITE 24	6.71	0.193333333	19.33333333	3.446329362	102.2438551
	6.5	0.84375	84.375	25.56894144	
	1.6	0.533333333	53.33333333	26.9368272	
	18899.47	75.59788	7559.788	45.81825818	
	125	0.625	62.5	0.473498916	
				102.2438551	

Table 5.6: water quality indices and water quality at different location

SITE NO	JANUARY			FEBRAURY		
	INDEX VALUE	WATER QUALITY STATUS	GRADING	INDEX VALUE	WATER QUALITY STATUS	GRADING
SITE 1	36.9943908	GOOD	B	33.93889698	GOOD	B
SITE 2	35.84170257	GOOD	B	30.14817236	GOOD	B
SITE 3	45.12094041	GOOD	B	37.64696416	GOOD	B
SITE 4	47.52487992	GOOD	B	23.74941323	EXCELLENT	A
SITE 5	45.51478059	GOOD	B	48.43851075	GOOD	B
SITE 6	52.08161066	POOR	C	45.73976252	GOOD	B
SITE 7	41.25872823	GOOD	B	39.28693168	GOOD	B
SITE 8	27.78734464	GOOD	B	27.25350822	GOOD	B
SITE 9	39.69108609	GOOD	B	42.91336641	GOOD	B
SITE 10	38.33472177	GOOD	B	30.3965507	GOOD	B

SITE 11	31.08868752	GOOD	B	30.45767353	GOOD	B
SITE 12	33.04930628	GOOD	B	32.56732226	GOOD	B
SITE 13	40.48321697	GOOD	B	26.28714068	GOOD	B
SITE 14	49.86835418	GOOD	B	51.06076631	POOR	C
SITE 15	37.41057865	GOOD	B	35.59065926	GOOD	B
SITE 16	34.42010953	GOOD	B	27.62735573	GOOD	B
SITE 17	40.94453588	GOOD	B	43.81617101	GOOD	B
SITE 18	71.35686542	POOR	C	74.21506878	POOR	C
SITE 19	41.87331769	GOOD	B	39.7180447	GOOD	B
SITE 20	32.5328144	GOOD	B	26.78699161	GOOD	B
SITE 21	43.1904295	GOOD	B	41.55287992	GOOD	B
SITE 22	68.65999188	POOR	C	75.91639213	VERY POOR	D
SITE 23	88.214841	VERY POOR	D	89.1558463	VERY POOR	D
SITE 24	94.56727968	VERY POOR	D	102.2438551	UNFIT	E

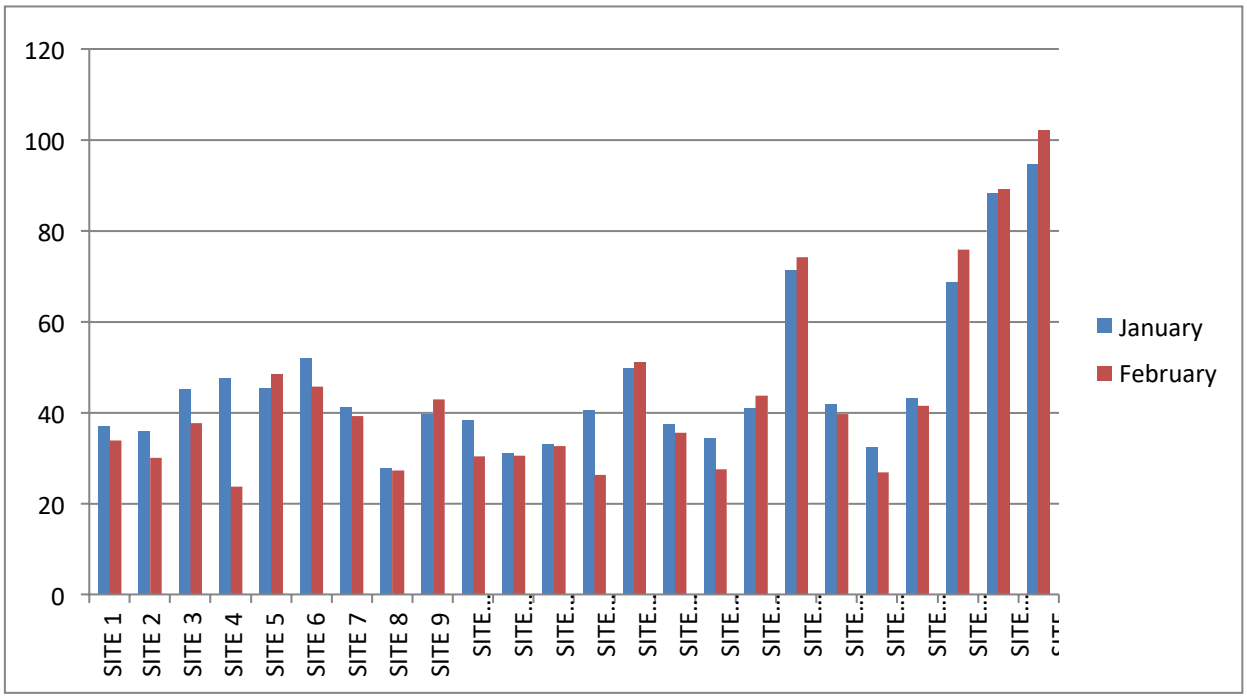

Fig no 5.1 WQI at various location along Mogral river

CHAPTER 6

CONCLUSION

In the present study, six different parameters are considered for the determination of water quality of Mogral river. Water samples were collected from twenty four sites and the parameter considered was pH, temperature, BOD, DO, chloride, alkalinity. Water quality index found using weighted arithmetic water quality index method.

The WQI for all samples found in the range of 27.79 – 94.56 in January and 23.75 – 102.24 in February. The water quality analysis shows that the 79.17% water samples were found as good category in January and 4.16% excellent category and 75% good category in February can be use for direct consumption while 12.5% belongs to poor category and 8.3% belongs to very poor category in January and 8.3% poor category and 8.3% very poor category in February shows that water is not suitable for direct consumption and also 4.16% belongs to unsuitable category in February which is unfit for drinking purpose and proper treatment required for any kind of usage. These variation in water quality might be due to high anthropogenic activities, illegal discharge of sewage, industrial effluent, lack of proper sanitation, unprotected river sites and urban runoff.

The concentration of chloride was found to be high in sites 22 – 24 due to intrusion of seawater or saline water which decline the quality of water. Open dumping sites like Chappady, PadyBellur, Alampady, Kopa and Patla tributary have presently no issues regarding their quality.

In order to increase the quality of Mogral river water necessary control or remedial measures have to be taken which includes regular and detailed water quality monitoring, provide fencing at locations of waste dumping sites from the bridges, silt removal and river cleaning, implementation and installation of sewage treatment plant and water drainage system, river bank protection works, construction of structure to prevent salt water intrusion into river at the downstream side, implement proper solid waste management system: collection, segregation, transportation, treatment, and disposal. Conduct public awareness programmes and install monitoring devices.

6.1 FUTURE SCOPE

- There is great scope for WQI as it can form the basis for identification of pollution hotspots and thus basis of evaluation of Goverenment interventions.
- To provide an overview and trend the water quality of Mogral river in Kasaragod city, which can be used for further development plans of the Mogral river.
- Helps in drawing comparison between water quality at different locations and thus reporting in an easily comprehendible manner.
- To give recommendations based on the study of the Mogral river stretch to water quality management authorities on how the result can be integrated for sustainable catchment management strategies of the basin.

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